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A nesting framework for Markov-switching GARCH modelling with an application to the German stock market

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In this paper, we establish a generalized two-regime Markov-switching GARCH model which enables us to specify complex (symmetric and asymmetric) GARCH equations that may differ considerably in their functional forms across the two Markov regimes. We show how previously proposed collapsing procedures for the Markov-switching GARCH model can be extended to estimate our general specification by means of classical maximum-likelihood methods. We estimate several variants of the generalized Markov-switching GARCH model using daily excess returns of the German stock market index DAX sampled during the last decade. Our empirical study has two major findings. First, our generalized model outperforms all nested specifications in terms of (a) statistical fit (when model selection is based on likelihood ratio tests) and (b) out-of-sample volatility forecasting performance. Second, we find significant Markov-switching structures in German stock market data, with substantially differing volatility equations across the regimes.

Keywords: Markov-switching models; GARCH models; Dynamics of stock-index returns; Volatility forecasting

JEL Classifications: C5, G10, G15

1. Introduction

Since the seminal papers of Engle (1982) and Bollerslev (1986) GARCH (generalized autoregressive conditional heteroscedasticity) models have become a standard tool in analysing the conditional variances of the returns from financial time series data. In order to cover specific volatility features like the well-known leverage effect and other volatility asymmetries in financial returns (e.g. Black 1976, Christie 1982, Schwert 1989), a plethora of GARCH specifications has been suggested in the literature, among the most prominent being the exponential GARCH (EGARCH) model introduced by Nelson (1991) and the threshold GARCH (TGARCH) model of Zakoïan (1994). Hentschel (1995) establishes a connection between many of these models by showing that their parametric forms are special cases of a Box–Cox (1964) transformation to the conditional standard deviation.

In contrast to these single-regime GARCH specifications, their multiple-regime Markov-switching counterparts are less straightforward to implement. An estimation problem typically inherent in Markov-switching GARCH models results from a phenomenon known as *path dependence* (Cai 1994, Hamilton and Susmel 1994). This dependence stems from the GARCH lag structure and, in a Markov-switching GARCH model, causes the regime-specific conditional variance to depend on the entire history of the data. Maximum likelihood estimation of path-dependent Markov-switching GARCH models turns out to be difficult, since the exact computation of the likelihood function is infeasible in practice. Recently, several authors have applied alternative methods to classical maximum likelihood techniques in order to estimate path-dependent Markov-switching GARCH specifications, such as a generalized method of moments procedure (Francq and Zakoïan 2008), Bayesian Markov chain Monte Carlo techniques (Bauwens *et al*. 2010, Bauwens *et al*. 2014) and a novel approach based on the Monte Carlo expectation–maximization algorithm and the Monte Carlo maximum-likelihood method (Augustyniak 2014).

Prior to these approaches to estimating path-dependent models, two modified Markov-switching GARCH specifications have been proposed in the literature, both circumventing the problem of path dependence and permitting direct

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estimation of all model parameters via classical maximum-likelihood techniques. First, Gray (1996) established a path-independent Markov-switching GARCH model which was refined by Klaassen (2002) and it is this flexible Gray–Klaassen Markov-switching framework that we expand in section 2. Compared to the above-mentioned methods for estimating path-dependent Markov-switching GARCH models, maximum-likelihood estimation within the Gray-Klaassen framework is easier to implement and requires less computational effort. As shown below, the Gray–Klaassen framework is appropriate for generalizing the regime-specific variance equations to more complex specifications than standard GARCH, which is our main justification for using this framework. A second specification avoiding path dependence is from Haas *et al.* (2004). Although this Markov-switching GARCH model has some advantages over the Gray-Klaassen framework with regard to the interpretation of the regime-specific volatility dynamics, we do not pursue this approach any further in this paper.

Markov-switching GARCH models are currently widely used in analysing the distribution of daily asset returns. Alexander (2008, p. 163) justifies their empirical implementation by stating that these models 'tell us a lot about volatility dynamics of equity markets and allow one to characterize its behaviour in [...] different market regimes', for example, during bull and bear market phases. Recent applications of Markov-switching GARCH models to commodity prices, exchange rate returns and stock-return data are presented, *inter alia*, in Alizadeh *et al.* (2008), Wilfling (2009) and Bohl *et al.* (2011). Apart from a few exceptions, the majority of two-regime Markov-switching GARCH specifications hitherto fitted to real-world data have one feature in common that appears unnecessarily restrictive. Despite the fact that the parameters in the variance equations are allowed to switch across both regimes, the overall functional forms of the two regime-specific GARCH equations are modelled as identical. The above-cited empirical applications, for example, all specify two-regime Markov-switching models with standard GARCH equations in each Markov-regime.†

In this paper, we develop a more flexible set-up by incorporating Hentschel's (1995) results on the nesting of distinct symmetric and asymmetric single-regime GARCH equations into the Gray-Klaassen Markov-switching GARCH framework. In this way, we establish a generalized regime-switching model that enables us to estimate complex GARCH equations of different functional forms across the Markov regimes. For example, our setup allows us to specify an EGARCH equation in Regime 1 and a standard GARCH specification in Regime 2.‡ In the empirical part of the paper, we apply our general Markov-switching GARCH approach to the excess returns generated by the German stock index DAX. Overall, we estimate several variants of our generalized Markov-switching GARCH model. (a) The most flexible variant which nests all other members of the Markov-switching GARCH family, and (b) alternative nested variants with regime-specific GARCH equations that have been widely used in previous empirical (single-regime) studies. Based on likelihood ratio tests, our most flexible Markov-switching GARCH specification is shown to outperform all nested variants in terms of statistical fit. We also find substantial improvements in this most flexible variant compared to all nested specifications (except for one) with respect to out-of-sample volatility forecasting performance.

The remainder of the paper is organized as follows. Section 2 formally establishes our generalized Markov-switching GARCH framework. For ease of readability, we derive the complete maximum-likelihood estimation procedure in the technical appendix to the paper. Section 3 describes the data-set and presents the estimation and volatility-forecasting results. The final section 4 summarizes the main findings and concludes.

2. A generalized Markov-switching GARCH model

In this section, we establish our generalized Markov-switching model, which enables us to specify and estimate GARCH equations of different functional forms in each of the distinct Markov regimes. To this end, we assume that the data-generating process (DGP) of the financial return r_t is affected by an unobserved latent random variable S_t representing the regime in which the DGP is operating at time t. For simplicity, we assume only the two distinct regimes 1 and 2 at any point in time. That is, we assume either $S_t = 1$ or $S_t = 2$ for all $t = 1, 2, \ldots$.

As the starting point of our derivation, we follow Hentschel's (1995) exposition and build up the so-called *Absolute Value GARCH* model for the return process $\{r_t\}$. However, we expand Hentschel's single-regime framework to a two-regime Markov-switching model. In order to do so, we let the return dynamics depend on the regime indicator $S_t = i, i = 1, 2$ and specify

$$r_{t+1} = \lambda_{S_t} + \gamma_{S_t}\sqrt{h_{S_t,t}} + \sqrt{h_{S_t,t}}\epsilon_{t+1}. \tag{1}$$

For ease of notation, from now on we write $\lambda_i = \lambda_{S_t}$, $\gamma_i = \gamma_{S_t}$ and $h_{i,t} = h_{S_t,t}$ when $S_t = i$. In equation (1), λ_i and γ_i are regime-specific constants, while $\{\epsilon_t\}$ denotes an i.i.d. process of standard normal variates. $h_{i,t}$ represents the conditional variance in Regime i, whose explicit modelling is treated below. The term $\lambda_i + \gamma_i\sqrt{h_{i,t}}$ on the right of equation (1) constitutes the mean equation of the return in Regime i and is known as the GARCH-in-Mean model. Our GARCH-in-Mean formulation in equation (1) constitutes the natural two-regime Markov-switching extension of the single-regime specification suggested by Campbell *et al.* (1997, p. 494).

†For recent extensions of Gray's (1996) Markov-switching framework to non-standard GARCH equations within the regimes, see Ané and Ureche-Rangau (2006) and Henry (2009), who consider Asymmetric Power GARCH (APGARCH) and EGARCH regimes as proposed in the single-regime setting by Ding *et al.* (1993) and Nelson (1991), respectively. In a similar vein, Haas (2010) and Ardia (2009) extend the above-mentioned alternative Markov-switching GARCH framework of Haas *et al.* (2004) by introducing regime-specific non-standard GARCH equations and by considering non-Gaussian innovations.

‡Lin (1998) also extends Gray's (1996) Markov-switching GARCH framework by incorporating Hentschel's (1995) single-regime GARCH specifications. In contrast to Lin (1998), we establish a rigorous fully fledged maximum-likelihood estimation framework with additional components (a GARCH-in-Mean term and time-varying transition probabilities) and integrate Klaassen's (2002) improvement.

The original GARCH-in-Mean model was introduced by Engle *et al.* (1987), and has since been used in many empirical studies on the behaviour of commodity prices and stock returns (e.g. Holt and Aradhyula 1998, Ghysels *et al.* 2005).

It should be noted that our specification in equation (1) differs slightly from the formulation used by many authors in the recent Markov-switching GARCH literature, with respect to the use of time indices. In equation (1), the return at time $t+1$ depends on the regime at time t (via S_t), whereas in the standard formulation, the return at time $t+1$ typically depends on the regime at time $t+1$. It is clearly possible to make our specification consistent with the standard literature by introducing the regime indicator $S_t^* = S_{t-1}$. However, in order (a) to consistently formulate all nested regime-specific model variants analogously to Hentschel's (1995) single-regime framework, and (b) to remain consistent with the single-regime GARCH-in-Mean specification suggested by Campbell *et al.* (1997, p. 494), we use the (non-standard) specification from equation (1) in this paper.

Based on the assumptions above, the conditional distribution of the return is a mixture of two normal distributions which can be written as

$$r_{t+1}|\phi_t \sim \begin{cases} N(\lambda_1 + \gamma_1\sqrt{h_{1,t}}, h_{1,t}) \text{ with probability } p_{1,t} \\ N(\lambda_2 + \gamma_2\sqrt{h_{2,t}}, h_{2,t}) \text{ with probability } (1-p_{1,t}) \end{cases}. \tag{2}$$

In equation (2), ϕ_t defines the information set as of date t and $p_{1,t} \equiv \Pr(S_t = 1|\phi_t)$ denotes the so-called *ex-ante* probability of being in Regime 1 at date t. Note that the information set ϕ_t basically coincides with the return path $\tilde{r}_t = \{r_t, r_{t-1}, \ldots\}$, but does not contain the path of the unobservable regime indicator S_t.

In modelling our regime-specific GARCH equations, we follow the path-independent methodology developed in Gray (1996). In order to circumvent the above-mentioned problem of path dependence, we specify the dynamics of the regime-specific conditional variance $h_{i,t}$ in terms of a lagged variance h_{t-1} and a shock term δ_t, both of which are appropriately weighted aggregates of the past conditional variances $h_{1,t-1}$ and $h_{2,t-1}$ from both Markov regimes. At this point, we use an econometric improvement of Gray's approach suggested by Klaassen (2002). His idea is to exploit all available information when integrating out the unobserved regimes in order to establish the aggregated variances and shock terms, whereas Gray uses only part of the information. More precisely, in specifying the volatility $h_{i,t}$ valid in Regime i, Klaassen computes the aggregated variance h_{t-1} and the shock terms δ_t on the basis of probabilities, which explicitly consider Regime i at time t. This modelling improvement is particularly efficient when the Markov-regimes appear to be highly persistent. In order to indicate the use of this additional information, we denote by $h_{t-1}^{(i)}$ the aggregated variance for date $t-1$, conditional on the fact that we are in Regime i on date t, and accordingly, we denote the shock terms by $\delta_t^{(i)}$. In particular, we specify both quantities as

$$\begin{aligned} h_{t-1}^{(i)} = {} & p_{1,t-1}^{(i)} \left(h_{1,t-1} + \left[\lambda_1 + \gamma_1 \sqrt{h_{1,t-1}} \right]^2 \right) \\ & + (1 - p_{1,t-1}^{(i)}) \left(h_{2,t-1} + \left[\lambda_2 + \gamma_2 \sqrt{h_{2,t-1}} \right]^2 \right) \\ & - \left[p_{1,t-1}^{(i)} \left(\lambda_1 + \gamma_1 \sqrt{h_{1,t-1}} \right) \right. \\ & \left. + (1 - p_{1,t-1}^{(i)}) \left(\lambda_2 + \gamma_2 \sqrt{h_{2,t-1}} \right) \right]^2 \end{aligned} \tag{3}$$

and

$$\begin{aligned} \delta_t^{(i)} = {} & p_{1,t-1}^{(i)} \frac{r_t - (\lambda_1 + \gamma_1 \sqrt{h_{1,t-1}})}{\sqrt{h_{1,t-1}}} \\ & + (1 - p_{1,t-1}^{(i)}) \frac{r_t - (\lambda_2 + \gamma_2 \sqrt{h_{2,t-1}})}{\sqrt{h_{2,t-1}}}, \end{aligned} \tag{4}$$

respectively, where the probabilities $p_{1,t-1}^{(i)}$ are calculated from equation (A12) in the appendix 1.†

Based on the aggregated variance and shock terms $h_{t-1}^{(i)}$ and $\delta_t^{(i)}$ from equations (3) and (4), we now define a preliminary two-regime conditional volatility equation as

$$\begin{aligned} \sqrt{h_{i,t}} &= \sqrt{\mathrm{Var}(r_{t+1}|\phi_t, S_t = i)} \\ &= \omega_i' + \alpha_i' \sqrt{h_{t-1}^{(i)}} |\delta_t^{(i)}| + \beta_i \sqrt{h_{t-1}^{(i)}}, \end{aligned} \tag{5}$$

where ω_i', α_i' and β_i denote regime-specific volatility parameters to be estimated from the data. It is obvious that the volatility equation (5) constitutes a standard GARCH(1,1) model in which the conditional variance terms and the shock terms have been replaced by the conditional standard deviations and the absolute shock terms, respectively. However, an important drawback of this volatility equation is that it cannot capture empirically well-documented (regime-specific) volatility asymmetries in financial returns (like the above-mentioned leverage effect). In order to resolve this deficit, we follow Hentschel (1995) and specify the second version of our two-regime conditional volatility equation as

$$\sqrt{h_{i,t}} = \omega_i' + \alpha_i' \sqrt{h_{t-1}^{(i)}} f_i(\delta_t^{(i)}) + \beta_i \sqrt{h_{t-1}^{(i)}} \tag{6}$$

with

$$f_i(\delta_t^{(i)}) = |\delta_t^{(i)} - b_i| - c_i(\delta_t^{(i)} - b_i), \tag{7}$$

where b_i, c_i represent regime-specific parameters. In what follows, we refer to the equations (6) and (7) as the *Absolute Value GARCH* (AVGARCH) model.

Although the AVGARCH specification is interesting in its own right, Hentschel (1995) demonstrates that a Box–Cox (1964) transformation of the conditional standard deviation in the equations (6) and (7) produces a rich class of models, including many well-known symmetric and asymmetric GARCH models as special cases. Adapting this approach to our two-regime Markov-switching framework by introducing the regime-specific parameters μ_i and ν_i, we transform our

†Instead of using the more informative Klaassen probabilities $p_{1,t-1}^{(i)}$, Gray (1996) uses the *ex-ante* probabilities $p_{1,t-1}$ from equation (2). This implies that Gray's aggregated variances and shock terms are equal, irrespective of the Markov-regime considered at date t.

Table 1. Nested GARCH models (within Regime i).

μ_i	ν_i	b_i	c_i	Model	Reference
0	1	0	Free	Exponential GARCH (EGARCH)	Nelson (1991)
1	1	0	$\lvert c_i\rvert \leq 1$	Threshold GARCH (TGARCH)	Zakoïan (1994)
1	1	Free	$\lvert c_i\rvert \leq 1$	Absolute Value GARCH (AVGARCH)	Hentschel (1995)
2	2	0	0	Standard GARCH (GARCH)	Bollerslev (1986)
2	2	Free	0	Nonlinear-asymmetric GARCH	Engle and Ng (1993)
2	2	0	Free	Glosten–Jagannathan–Runkle GARCH	Glosten *et al.* (1993)
Free	μ_i	0	0	Nonlinear ARCH	Higgins and Bera (1992)
Free	μ_i	0	$\lvert c_i\rvert \leq 1$	Asymmetric power ARCH	Ding *et al.* (1993)

Note: Table compiled from Hentschel (1995, table 1).

conditional volatility equation (6) to

$$\frac{\sqrt{h_{i,t}}^{\mu_i} - 1}{\mu_i} = \omega_i + \alpha_i \sqrt{h_{t-1}^{(i)}}^{\mu_i} f_i^{\nu_i}(\delta_t^{(i)}) + \beta_i \frac{\sqrt{h_{t-1}^{(i)}}^{\mu_i} - 1}{\mu_i}. \tag{8}$$

The parameters ω_i and α_i are reparameterizations of ω_i' and α_i' in equation (6), while μ_i determines the shape of the Box–Cox transformation in Regime i. For $0 \leq \mu_i \leq 1$, the transformation of the conditional standard deviation $\sqrt{h_{i,t}}$ is concave, while it is convex for $\mu_i > 1$. The parameter ν_i transforms the regime-specific function $f_i(\cdot)$. For $0 < \nu_i < 1$, the function $f^{\nu_i}(\cdot)$ becomes concave on either side of b_i, while it becomes convex for $\nu_i > 1$. Table 1, compiled from table 1 in Hentschel (1995, p. 79), reveals how our volatility equation (8) for Regime i nests many GARCH models scattered throughout the literature, by imposing appropriate restrictions on the parameters μ_i, ν_i, b_i and c_i.

All models presented in table 1 should ensure non-negativity of the conditional variances within each regime, that is $h_{i,t} \geq 0$. In the case of Bollerslev's (1986) standard GARCH(1,1) model, this requirement implies $\omega_i > 0, \alpha_i, \beta_i \geq 0$. However, while these conditions are sufficient, they are not always necessary for ensuring positive conditional variances in other specifications from table 1. For example, the EGARCH model does not impose any sign restrictions on the parameters ω_i, α_i, and on β_i to ensure positive conditional variances. A convenient choice of the parameter c_i on the right-hand side of equation (7) is $\lvert c_i\rvert \leq 1$, since this condition ensures a positive value of $f^{\nu_i}(\delta_t^{(i)})$. However, $\lvert c_i\rvert \leq 1$ is neither a necessary nor a sufficient condition to ensure $h_{i,t} \geq 0$. For a detailed analysis of regime-specific parameter restrictions ensuring positive conditional variances, see Hentschel (1995, appendix A).

Finally, we close our econometric model by specifying the probabilistic nature of the regime indicator S_t. In our study, we let $\{S_t\}$ follow a two-state first-order Markov process with time-varying transition probabilities and write this as

$$\begin{aligned} \Pr(S_t = 1 | S_{t-1} = 1, r_t) &= P_t, \\ \Pr(S_t = 2 | S_{t-1} = 1, r_t) &= 1 - P_t, \\ \Pr(S_t = 1 | S_{t-1} = 2, r_t) &= 1 - Q_t, \\ \Pr(S_t = 2 | S_{t-1} = 2, r_t) &= Q_t. \end{aligned} \tag{9}$$

The probability of being in Regime i for $i = 1, 2$ depends on realizations in $\tilde{r}_t$ and $\{S_t\}$ only through S_{t-1}. For the time-varying transition probabilities, we assume

$$\begin{aligned} P_t &= \Phi(d_1 + e_1 \cdot r_t), \\ Q_t &= \Phi(d_2 + e_2 \cdot r_t), \end{aligned} \tag{10}$$

with $\Phi(\cdot)$ denoting the cumulative distribution function of a standard normal variate and d_1, d_2, e_1, e_2 representing parameters to be estimated from the data.†

Our Markov-switching GARCH model established in equations (1)–(10) can now be estimated using classical maximum likelihood techniques. The log-likelihood function is constructed recursively and we present its exact form in equations (A1)–(A13) of the appendix 1. In the next section, we apply this generalized Markov-switching GARCH framework to the daily excess returns of the German stock index DAX during the last decade.

3. Empirical application

3.1. Data

In this section, we analyse the Markov-switching mean and volatility structures inherent in the daily excess returns of the German stock market between 3 January 2000 and 31 December 2009 (2554 observations). We calculate the excess returns r_t by subtracting an appropriately defined risk-free interest rate from the returns of the German stock index DAX.‡ Our DAX returns used for calculating the excess returns are adjusted for dividend payments. As the risk-free interest rate, we use the *Euro OverNight Index Average* (EONIA), which we convert into daily returns by dividing the given annualized EONIA rate by 250.§

Figure 1 displays the German stock index DAX (upper panel) and the corresponding DAX excess returns r_t (lower panel) during the sampling period. The trajectory of excess returns clearly exhibits the two most prominent features that are well documented in the financial literature on asset-return dynamics, namely volatility clustering and a time-varying mean.

†Our specification (10) of the time-varying transition probabilities stems from Gray (1996). For an economic justification, see Diebold *et al.* (1994) and the literature cited therein.
‡Our interest-rate data is provided by the *Deutsche Bundesbank*, and the stock-market data is from *Datastream* (daily closing prices).
§We divide by 250 in order to be consistent with the approximate number of observations per year available for DAX returns.

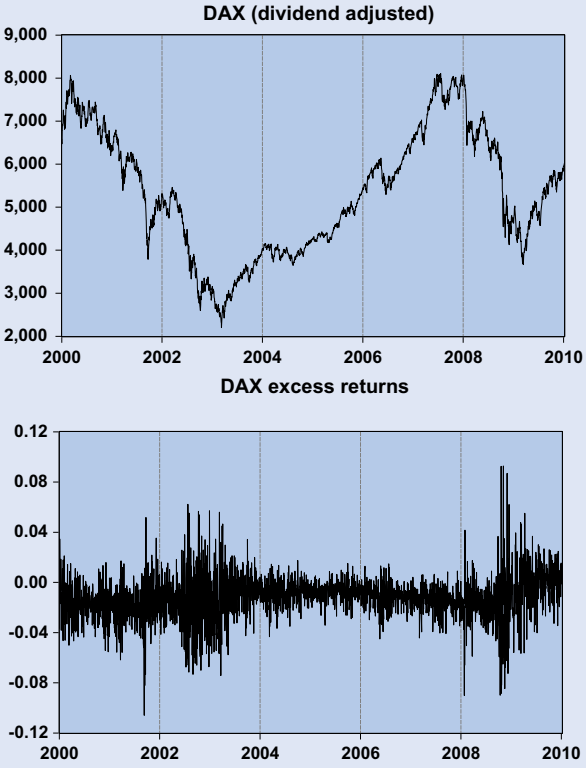


Figure 1. The upper panel displays the daily German stock index DAX between 3 January 2000 and 31 December 2009 (2554 observations). The lower panel contains the corresponding daily excess returns (adjusted for dividend payments) during the sampling period. As the risk-free interest rate, we use the *Euro Overnight Index Average* (EONIA). The trajectory of excess returns exhibits volatility clustering and a time-varying mean.

3.2. Estimation results

We now analyse the dynamic structures of the DAX excess returns r_t within the Markov-switching GARCH framework developed in section 2.

3.2.1. Technical issues. Tables 2–4 below display the maximum-likelihood (ML) estimation results for several single-regime GARCH and two-regime Markov-switching GARCH models represented by equations (1)–(10). We numerically maximized the log-likelihood functions from equations (A1) to (A13) using the BFGS-algorithm from the FMINCON module of the software package MATLAB.

Since the log-likelihood functions of Markov-switching GARCH models are prone to multiple local optima, a good choice of starting values is crucial to the optimization process. To this end, we followed the recommendation of Gilli and Schumann (2010) and implemented a best-of-N-restarts strategy with random sampling points. More specifically, we first constrained all the parameters to feasible ranges. In a second step, we randomly drew 20 000 parameter combinations from these feasible ranges, computed the corresponding log-likelihood values, and saved the $N = 40$ best of these parameter combinations. We then selected these best combinations as the distinct starting values (vectors) in our process of maximizing the value of the log-likelihood function. To circumvent numerical problems stemming from the absolute value function appearing on the right of equation (7), we replaced the argument of the absolute value function with the hyperbolic approximation proposed by Hentschel (1995). Standard errors were computed from the diagonal of the heteroscedasticity-consistent (White-robust) covariance matrix.

Our Markov-switching GARCH framework enables us to estimate a plethora of distinct two-regime Markov-switching GARCH models. Restrictions on the regime-dependent parameters μ_i, ν_i, b_i and c_i lead to specific functional forms of the two variance equations, for example, to an EGARCH equation in Regime 1 ($\mu_1 = 0, \nu_1 = 1, b_1 = 0, c_1 =$ free) and a standard GARCH equation in Regime 2 ($\mu_2 = 2$, $\nu_2 = 2$, $b_2 = 0$, $c_2 = 0$). In what follows, we refer to this latter specification as a Markov-switching EGARCH–GARCH model and, based on the terminology in table 1, we analogously use the phrasing TGARCH–GARCH, EGARCH–EGARCH and so on for our two-regime models. We refer to the two-regime Markov-switching GARCH model without any restrictions on the parameters μ_i, ν_i, b_i and c_i as the Free–Free model. Finally, we refer to all single-regime models in the standard way (GARCH, EGARCH, Free, and so on).

3.2.2. Preliminaries. Our model set-up in section 2 includes the GARCH-in-Mean term from equation (1) and the time-varying transition probabilities specified in the equations (9) and (10). These two components are generally discussed separately from each other in the single-regime and Markov-switching GARCH literature. In the first empirical step, our objective is to investigate which of these two extensions makes a major contribution when fitting Markov-switching GARCH models to our DAX data. For this purpose, we estimated four variants of the Free–Free model.

(a) A full variant including both, the GARCH-in-Mean term ($\gamma_1, \gamma_2 \neq 0$) and time-varying transition probabilities ($e_1, e_2 \neq 0$).
(b) A variant with GARCH-in-Mean ($\gamma_1, \gamma_2 \neq 0$) with constant transition probabilities ($e_1, e_2 = 0$).
(c) A variant without the GARCH-in-Mean term ($\gamma_1, \gamma_2 = 0$) with time-varying transition probabilities ($e_1, e_2 \neq 0$).
(d) A variant without the GARCH-in-Mean term ($\gamma_1, \gamma_2 = 0$) with constant transition probabilities ($e_1, e_2 = 0$).

Table 2 reports likelihood-based goodness-of-fit statistics enabling us to compare the four specifications with each other. Denoting the number of parameters of a given variant by K, and the maximized log-likelihood value by ℓ, we computed the Bayesian information criterion as BIC $= -2 \cdot \ell + K \cdot \log(T)$. According to the BIC, the variant (b) with GARCH-in-Mean term and constant transition probabilities provides the best data fit. This result is further strengthened by the three likelihood ratio tests (LRT) of the respective problems 'H_0: restricted Free–Free model (variants (b), (c), (d)) versus H_1: unrestricted Free–Free model (variant (a))'. Clearly, the full variant (a) outperforms the nested variants (c) and (d) at all conventional significance levels. By contrast, it does not outperform the nested variant (b) at the 5% level. Thus, the BIC

Table 2. Likelihood-based goodness-of-fit: alternative restrictions of the Free–Free model.

Component	K	ℓ	BIC		LRT	p-value
			Value	Rank		
GARCH-in-Mean: $(\gamma_1, \gamma_2 \neq 0)$ Time-varying Trans. prob. $(e_1, e_2 \neq 0)$	22	7314.0078	−14455.4165	2		
Constant Trans. prob. $(e_1, e_2 = 0)$	20	7311.1366	−14465.3649	1	5.7424*	0.0566
No GARCH-in-Mean: $(\gamma_1, \gamma_2 = 0)$ Time-varying Trans. prob. $(e_1, e_2 \neq 0)$	20	7218.8189	−14280.7295	4	190.3778***	0.0000
Constant Trans. prob. $(e_1, e_2 = 0)$	18	7213.0827	−14284.9479	3	201.8502***	0.0000

Notes: The unrestricted Free–Free model with all components (with GARCH-in-Mean and time-varying transition probabilities) is given in equations (1)–(10). Omitting the GARCH-in-Mean component and/or specifying constant transition probabilities impose $\gamma_1, \gamma_2 = 0$ and/or $e_1, e_2 = 0$. K denotes the number of parameters of a given specification. ℓ is the maximized log-likelihood value. BIC is the Bayesian information criterion computed as $\text{BIC} = -2 \cdot \ell + K \cdot \log(T)$ with $T = 2554$ representing the sample size. LRT is the likelihood ratio test statistic of the problem 'H_0: restricted Free–Free model versus H_1: unrestricted Free–Free model' computed as twice the difference between the respective ℓ-values. The LRT statistics are asymptotically χ^2-distributed under the respective null hypotheses with degrees of freedom equal to the difference in the number of parameters (K) between the unrestricted Free–Free model ($K = 22$) and the respective restricted specifications from the null hypothesis ($K = 20$ or $K = 18$).
*,** and *** denote statistical significance at 10%, 5% and 1% levels, respectively.

ranking and this latter LRT result suggest a need to specify our Markov-switching GARCH models with a GARCH-in-Mean term ($\gamma_1, \gamma_2 \neq 0$) and with constant transition probabilities ($e_1, e_2 = 0$). According to equations (9) and (10), the constant transition probabilities are defined as $P = \Phi(d_1)$ and $Q = \Phi(d_2)$.

3.2.3. Model comparisons and specification issues. In the empirical analysis, we confine ourselves to estimating five distinct two-regime Markov-switching GARCH specifications for the DAX excess returns, namely the (a) standard GARCH–GARCH model, (b) AVGARCH–AVGARCH model, (c) EGARCH–GARCH model, (d) EGARCH–EGARCH model, and (e) the Free–Free model without any restrictions on the parameters μ_i, ν_i, b_i, c_i. Table 3 reports likelihood-based model comparisons of this basic group with alternative specifications. We also estimated all five two-regime Markov-switching GARCH models without the GARCH-in-Mean component and conducted likelihood ratio tests in order to assess the statistical benefit of including the GARCH-in-Mean term. In table 3, the corresponding LRT statistics are reported in group (2). Obviously, all two-regime Markov-switching GARCH models with the GARCH-in-Mean component outperform their two-regime counterpart specification without the GARCH-in-Mean term at all conventional levels. For the purposes of comparison, we also estimated four single-regime specifications, all with and without GARCH-in-Mean terms. Again, the LRTs reported in group (4) indicate the additional value of the GARCH-in-Mean component for all single-regime models (at the 5% level for the standard GARCH model and far below the 1% level for all other specifications).

An important specification issue concerns the number of Markov-regimes in our regime-switching representation (1)–(10). Testing the significance of a second Markov-regime is not an easy task, due to a potential identification issue known as the Davies Problem (see Davies 1977). The identification problem implies that the conventional LRT may be statistically improper, since we can no longer assume the validity of the χ^2-approximation to the LRT statistic under the null hypothesis of a single Markov-regime. However, Gelman and Wilfling (2009) assess the finite-sample properties of the conventional LRT statistic (defined as twice the difference between the log-likelihoods of the two-regime Markov-switching and the single-regime specifications) for a GARCH–GARCH model by means of a parametric bootstrapping procedure. Their results indicate that the null distribution of the LRT statistic typically does not exhibit large deviations from the χ^2-distribution with degrees of freedom equal to the difference in the number of parameters between the two-regime and the single-regime specifications. Encouraged by these simulation results, we computed conventional LRT statistics for comparing single-regime GARCH, AVGARCH, EGARCH, Free specifications with their two-regime counterpart GARCH–GARCH, AVGARCH–AVGARCH, EGARCH–EGARCH and Free–Free specifications. All these LRT statistics, displayed in group (3) of table 3, are so extreme that they exceed all critical values used in practice, thus endorsing two-regime specifications.

Next, we address the question as to which of our five alternative Markov-switching GARCH specifications (including a GARCH-in-Mean term) provide the best data fit. Obviously, we cannot compare all models against each other on the basis of likelihood ratio tests, since two distinct specifications

Table 3. Likelihood-based model comparisons (two-regime and single-regime GARCH specifications).

Model	K	ℓ	BIC		LRT	p-value
			Value	Rank		
(1) GARCH-in-Mean ($\gamma_1, \gamma_2 \neq 0$):						
GARCH-GARCH	12	7244.5063	−14394.8676	5	133.2606***	0.0000
AVGARCH-AVGARCH	16	7304.2539	−14482.9811	2	13.7654***	0.0081
EGARCH-GARCH	13	7284.7411	−14467.4918	3	52.7910***	0.0000
EGARCH-EGARCH	14	7301.4672	−14493.0986	1	19.3388***	0.0036
Free-Free	20	7311.1366	−14465.3649	4		
(2) No GARCH-in-Mean ($\gamma_1, \gamma_2 = 0$):						
GARCH-GARCH	10	7190.1711	−14301.8880	12	108.6704***	0.0000
AVGARCH-AVGARCH	14	7206.7539	−14303.6720	11	195.0000***	0.0000
EGARCH-GARCH	11	7197.2110	−14308.1224	10	175.0602***	0.0000
EGARCH-EGARCH	12	7206.8472	−14319.5494	7	189.2400***	0.0000
Free-Free	18	7213.0827	−14284.9479	13	196.1078***	0.0000
(3) GARCH-in-Mean ($\gamma \neq 0$):						
GARCH	5	7150.5831	−14261.9391	18	187.8464	
AVGARCH	7	7185.6363	−14316.3547	8	237.2352	
EGARCH	6	7184.8523	−14322.6321	6	233.2298	
Free	9	7189.4511	−14308.2935	9	243.3709	
(4) No GARCH-in-Mean ($\gamma = 0$):						
GARCH	4	7147.7305	−14264.0793	17	5.7052**	0.0169
AVGARCH	6	7161.6314	−14276.1902	15	48.0099***	0.0000
EGARCH	5	7160.5020	−14281.7768	14	48.7007***	0.0000
Free	8	7164.9178	−14267.0722	16	49.0667***	0.0000

Notes: All two-regime models were estimated with constant transition probabilities ($e_1, e_2 = 0$). K denotes the number of parameters, ℓ the maximized log-likelihood value of a given model. BIC is the Bayesian information criterion computed as BIC $= -2 \cdot \ell + K \cdot \log(T)$ with $T = 2554$. LRT denotes the conventional likelihood ratio test statistic for the following respective testing problems. (1) Two-regime models, GARCH-in-Mean: 'H_0: the considered two-regime specification versus H_1: the Free–Free specification with GARCH-in-Mean'. (2) Two-regime models, no GARCH-in-Mean: 'H_0: the considered two-regime specification versus H_1: the corresponding two-regime counterpart specification with GARCH-in-Mean'. (3) Single-regime models, GARCH-in-Mean: 'H_0: the considered single-regime specification versus H_1: the corresponding two-regime counterpart specification with GARCH-in-Mean'. (4) Single-regime models, no GARCH-in-Mean: 'H_0: the considered single-regime specification versus H_1: the corresponding single-regime counterpart specification with GARCH-in-Mean'. Except for testing problem (3), the LRT statistics are asymptotically χ^2-distributed under the respective null hypotheses with the degree of freedom equal to the difference in the number of parameters between the H_1- and the H_0-specifications. For the testing problem (3), the null distribution of the LRT statistic is unknown. *,** and *** denote statistical significance at 10, 5 and 1% levels, respectively.

need to be nested in order to ensure that the LRT is valid. However, since our Markov-switching Free–Free model nests all other specifications (see table 1), we restrict our attention to the four LR testing problems (a) 'H_0: GARCH–GARCH versus H_1: Free-Free', (b) 'H_0: AVGARCH–AVGARCH versus H_1: Free-Free', (c) 'H_0: EGARCH–GARCH versus H_1: Free-Free' and (d) 'H_0: EGARCH–EGARCH versus H_1: Free-Free'. The LRTs, displayed in group (1) of table 3, reject all specifications in favor of the Free–Free model at the 1% significance level. Additionally, we also provide values of the Bayesian Information Criterion (BIC) so as to rank all specifications shown in table 3 (i.e. also the non-nested two-regime Markov-switching GARCH models, as well as the single-regime variants) against each other. The BIC clearly ranks the five two-regime Markov-switching GARCH models with the GARCH-in-Mean term being the best among all specifications considered. However, the BIC ranks the Free–Free-GARCH-in-Mean model only fourth, the two best ranked models being the more parsimonious EGARCH–EGARCH and AVGARCH–AVGARCH specifications. Interestingly, the single-regime EGARCH model with the GARCH-in-Mean term is ranked sixth, according to the BIC, thus beating all two-regime Markov-switching GARCH models without the GARCH-in-Mean term.

In view of these model comparisons, we decided to present detailed ML estimation results of the four best two-regime Markov-switching GARCH specifications with the GARCH-in-Mean and, additionally, for the single-regime EGARCH model with the GARCH-in-Mean term in table 4. To assess the statistical significance of the model parameters, we consider the conventional t-statistic whose exact finite-sample distribution is generally unknown in our estimation set-up. However, we can make an asymptotic inference by noting (a) that our ML estimators are asymptotically normally distributed, and (b) that our standard errors constitute (weakly) consistent estimates of the true standard deviations of the ML estimators.†

3.2.4. Volatility and regime-persistence results. An important econometric issue concerns the persistence of volatility shocks. In a standard single-regime GARCH(1,1)-equation of the form $h_t = \omega + \alpha \cdot h_{t-1}\delta_t^2 + \beta h_{t-1}$, the persistence of volatility shocks is typically measured by the sum $\alpha + \beta$. The

†Consequently, under the null hypothesis of a single parameter being equal to 0, our t-statistics should converge in distribution towards a standard normal variate, implying critical values of 1.6449, 1.9600 and 2.5758 at the 10, 5 and 1% levels, respectively, for the absolute value of the t-statistic (see Greene 2008, appendix D).

Table 4. Estimates of alternative single-regime GARCH and two-regime Markov-switching GARCH specifications.

	EGARCH	AVGARCH–AVGARCH	EGARCH–GARCH	EGARCH–EGARCH	Free–Free
μ_1	0	1	0	0	0.9696***
					(0.0049)
μ_2		1	2	0	0.5403***
					(0.1986)
ν_1	1	1	1	1	0.9857***
					(0.2961)
ν_2		1	2	1	0.8329
					(0.7823)
λ_1	−0.0061***	0.0036	-5.6539×10^{-5}	3.6845×10^{-5}	0.0038***
	(0.0001)	(0.0031)	(0.0002)	(0.0007)	(0.0002)
λ_2		0.0049	0.0008	0.0112***	0.0070***
		(0.0035)	(0.0006)	(0.0025)	(0.0011)
γ_1	−0.3747***	−1.5505***	−1.1685***	−1.1650***	−1.5668***
	(0.0001)	(0.2941)	(0.0112)	(0.1935)	(0.0525)
γ_2		−0.6448***	−0.3646***	−0.8866***	−0.7473***
		(0.1770)	(0.0108)	(0.0729)	(0.0136)
ω_1	−0.0997***	0.0019	−0.7323***	−0.7466***	0.0022***
	(2.9568×10^{-5})	(0.0012)	(0.0111)	(0.0850)	(0.0001)
ω_2		0.0028	$1.2332 \times 10^{-5*}$	−0.9229***	0.0212
		(0.0018)	(6.9006×10^{-6})	(0.0658)	(0.0172)
α_1	0.0730***	0.0094***	0.0211***	0.0205**	0.0095***
	(0.0001)	(0.0028)	(0.0077)	(0.0103)	(0.0027)
α_2		0.0394***	0.0493***	0.0083**	0.0337***
		(0.0135)	(0.0140)	(0.0034)	(0.0109)
β_1	0.9899***	0.3935***	0.9856***	0.9800***	0.3991***
	(0.0003)	(0.0560)	(0.0061)	(0.0263)	(0.0088)
β_2		0.6898***	0.7472***	0.7864***	0.7249***
		(0.0625)	(0.0203)	(0.0048)	(0.0810)
d_1		2.6687***	2.7890***	2.8194***	2.6711***
		(0.4034)	(0.1824)	(0.6080)	(0.4394)
d_2		2.4786***	2.5687***	2.5936***	2.4791***
		(0.2492)	(0.1879)	(0.2365)	(0.2374)
$P = \Phi(d_1)$		0.9962***	0.9974***	0.9976***	0.9962***
		(0.0046)	(0.0015)	(0.0046)	(0.0049)
$Q = \Phi(d_2)$		0.9934***	0.9949***	0.9953***	0.9934***
		(0.0046)	(0.0028)	(0.0033)	(0.0044)
b_1	0	1.0882	0	0	1.0737
		(0.7418)			(0.6672)
b_2		0.8909	0	0	1.0843***
		(0.6518)			(0.0958)
c_1	0.3648***	1.0000	2.5930***	2.3202	0.9883*
	(0.0004)	(5.3968)	(0.7075)	(10.1624)	(0.5218)
c_2		0.4295	0	5.3058	0.0812**
		(4.4067)		(36.9273)	(0.0384)

Notes: Estimates for parameters from equations (1) to (10). Standard errors are in parentheses. For the single-regime EGARCH model, all parameters are indexed by the subscript '1'. *,** and *** denote statistical significance at 10, 5 and 1% levels, respectively.

higher the value of $\alpha + \beta$, the longer it takes until a volatility shock dies out. In particular, when $\alpha + \beta = 1$, volatility shocks have a permanent effect and the unconditional variance of the process becomes infinitely large. In view of these considerations within a single-regime framework, it appears reasonable to measure the persistence of volatility shocks in a two-regime Markov-switching GARCH(1,1) model by the regime-specific sums $\alpha_i + \beta_i$ for $i = 1, 2$. However, the situation turns out to be more complicated, since in general, it is the interaction between the regime-specific volatility parameters and the transition probabilities of the regime indicator S_t, which determines the variance-stability of a Markov-switching GARCH model. Abramson and Cohen (2007) state the necessary and sufficient conditions for a standard two-regime Markov-switching GARCH-GARCH model to be *asymptotically wide-sense stationary*, and therefore to possess finite second-order moments. However, analogous results for any other of our two-regime Markov-switching GARCH specifications are currently not available in the literature and their derivation constitutes a substantial task for future research.

In applied work, it is common practice to restrict oneself to analysing the persistence of volatility shocks separately within each Markov regime. From the ML estimates of our two-regime Markov-switching GARCH–GARCH model (for reasons of brevity not presented in table 4), we find that the respective regime-specific sums $\hat{\alpha}_i + \hat{\beta}_i$ are given by 0.6753 and 0.2074, indicating covariance stationarity with rather low and moderate degrees of volatility persistence in both Markov-regimes. A similar result holds for Regime 2 of our Markov-switching EGARCH–GARCH model (Column 3 of

Table 5. Diagnostics for residual normality and independence.

	EGARCH	AVGARCH–AVGARCH	EGARCH–GARCH	EGARCH–EGARCH	Free–Free
JB	64.5313***	6.2189**	57.6936***	16.7150***	5.9318*
	(0.0000)	(0.0446)	(0.0000)	(0.0002)	(0.0515)
KS	0.0127	0.0268*	0.0133	0.0228	0.0197
	(0.7997)	(0.0503)	(0.7558)	(0.1391)	(0.2707)
AD	1.6346	2.3136*	0.8692	1.7285	1.0934
	(0.1474)	(0.0622)	(0.4337)	(0.1302)	(0.3118)
LB_1	0.8702	1.0585	1.9885	1.1249	1.9564
	(0.3509)	(0.3036)	(0.1585)	(0.2889)	(0.1619)
LB_2	3.1857	2.8322	2.7711	2.0871	2.0830
	(0.2033)	(0.2427)	(0.2502)	(0.3522)	(0.3529)
LB_3	10.1415**	2.8479	6.3600*	2.1317	2.9152
	(0.0174)	(0.4157)	(0.0954)	(0.5455)	(0.4049)
LB_5	10.6526	8.0016	11.7808**	8.8844	4.5217
	(0.0587)*	(0.1561)	(0.0379)	(0.1138)	(0.4770)
LB_{10}	12.1236	11.3766	22.8869**	10.8686	7.4162
	(0.2769)	(0.3289)	(0.0112)	(0.3678)	(0.6857)
LB_{15}	15.8635	16.8867	25.1543**	14.9019	13.1489
	(0.3912)	(0.3257)	(0.0479)	(0.4585)	(0.5908)

Notes: JB, KS, AD denote the Jarque-Bera, Kolmogorov-Smirnov and Anderson-Darling test statistics for residual normality, respectively. LB_i denotes the Ljung-Box-Q statistic for serial correlation of the residuals out to lag i. p-values are in parentheses. *,** and *** denote statistical significance at 10, 5 and 1% levels, respectively.

table 4), for which we find $\hat{\alpha}_2 + \hat{\beta}_2 = 0.7965$. For the most general Markov-switching Free–Free model, a sufficient condition for covariance stationarity in Regime i is given by

$$E\left[\left[\alpha_i \cdot \mu_i \cdot f^{\nu_i}(\epsilon_t) + \beta_i\right]^{2/\mu_i}\right] < 1 \tag{11}$$

(see Nelson 1990). Hentschel (1995) shows that for an AVGARCH specification with $\mu_i = \nu_i = 1$, condition (11) is equivalent to

$$\begin{aligned}&\alpha_i^2(1+b_i^2)(1+c_i^2) + \beta_i^2 + 2\alpha_i\beta_i b_i c_i \\ &\quad + 4\alpha_i(\beta_i + \alpha_i b_i c_i)\phi(b_i) \\ &\quad + 2\alpha_i(\beta_i b_i + \alpha_i(1+b_i^2)c_i)(2\Phi(b_i) - 1) < 1,\end{aligned} \tag{12}$$

with $\phi(\cdot)$ and $\Phi(\cdot)$ denoting the probability density and cumulative distribution functions of the standard normal distribution, while for a regime-specific EGARCH equation, condition (11) converges to

$$\beta_i < 1. \tag{13}$$

For both regimes in our Markov-switching AVGARCH–AVGARCH specification, the estimates from Column 2 of table 4 yield the values 0.1727 and 0.5615 when inserted into the left-hand side of condition (12), thus again indicating covariance stationarity with low and moderate degrees of volatility persistence in both Markov-regimes.

For the three EGARCH regimes in our EGARCH–GARCH and EGARCH–EGARCH specifications, the estimates of the parameters β_1 and β_2 are given by 0.9856, 0.9800 and 0.7864, indicating covariance stationarity but, at least in two regimes, with relatively high degrees of volatility persistence. For the Markov-switching Free–Free specification, no closed-form solution to the expectation on the left of Condition (11) exists. However, we calculated the two regime-specific expectations by numerical integration and obtained the values 0.4221 and 0.6863, indicating covariance-stationary Markov-regimes with moderate volatility persistence.

The constant transition probabilities $P = \Phi(d_1)$ and $Q = \Phi(d_2)$ specified in equations (9) and (10) represent the likelihood that no switch in the two Markov-regimes occurs between $t-1$ and t. Evidently, these staying probabilities are above 0.99 for all of our four two-regime specifications. To gain a better understanding of the amount of regime persistence, we can consider the expected durations $(1-P)^{-1}$ and $(1-Q)^{-1}$ of the respective regimes. For the Free–Free model, for instance, these expected durations are given by 263.2 and 151.5 (trading) days.

3.2.5. Residual diagnostics. Our approach to calculating residuals is based on the Rosenblatt (1952) transform and follows the methodology summarized by Haas (2010). To outline the procedure, we denote the theoretical conditional cumulative distribution function of r_{t+1} given by equation (2) as $F(r_{t+1}|\phi_t)$. Furthermore, we let $\hat{F}(r_{t+1}|\phi_t)$ denote the fitted cumulative distribution function evaluated at the observed value of the return at time $t+1$ and at the ML parameter estimates. We now define the residual at time $t+1$ as

$$\hat{\epsilon}_{t+1} = \Phi^{-1}\left(\hat{F}(r_{t+1}|\phi_t)\right), \tag{14}$$

where $\Phi^{-1}(\cdot)$ is the quantile function of the standard normal distribution. If the model is correctly specified, the residuals defined in equation (14) are realizations of an i.i.d. standard normal process.

To check for (standard) normality, we apply the well-known moment-based Jarque-Bera (JB) test, as well as the Kolmogorov-Smirnov (KS) and the Anderson-Darling (AD) empirical distribution tests.† While the KS and AD tests in table 5 do not reveal significant deviations from the normal distribution at the 5% level for any of our specifications, the moment-based JB test indicates substantial deviations for the majority of the models. Only for the Free–Free and the AVGARCH–AVGARCH models does the JB test not reject the null hypothesis of a normal distribution at the 5% and

†For details on these tests, see Haas (2010) and the cited literature.

1% levels, respectively. The lower part of table 5 displays the results of Ljung-Box-Q tests for serial correlation among the residuals for various lags. Except for the single-regime EGARCH and the EGARCH–GARCH model, the tests do not reveal evidence supporting the existence of autocorrelation in the residuals.

3.2.6. Conditional variances and regime probabilities. Figure 2 displays the conditional variances and regime-1 probabilities of the daily excess returns of the German stock market index DAX, as estimated for our five specifications from table 4. For each two-regime model, we depict two distinct regime-1 probabilities, namely the *ex-ante regime-1 probabilities* (thin lines in the left panels) along with the so-called *smoothed regime-1 probabilities* (bold lines in the left panels). We derived the *ex-ante* regime-1 probabilities $p_{1,t} = \Pr(S_t = 1|\phi_t)$ as a by-product of our estimation procedure according to equation (A7) in the appendix 1. Unlike the *ex-ante* probabilities, the smoothed regime-1 probabilities $\Pr(S_t = 1|\phi_T)$ are based on the information from the full sample up to time T. We computed the smoothed regime-1 probabilities according to the algorithm described in the appendix of Hamilton (1994, Chapter 22).

For all five models, the conditional variances exhibit a strikingly uniform pattern during the sampling interval between 2000 and 2010. The beginning of the decade started with a period of relatively high volatility in the German stock market, with a pronounced peak in conditional variances around 11th September 2001. After a short phase of normalization, an extended period of high stock-market volatility occurred between mid-2002 and mid-2003, reflecting the German bear market in which the DAX fell from about 5000 to 2000 index points. Between 2004 and the beginning of 2008, the conditional volatility of the DAX was comparably low. This period of low market fluctuation came to an abrupt end at the beginning of 2008, when the German stock market began to respond to the subprime crisis by plummeting stock prices. However, the highest volatility peak occurred around 15 September 2008, when Lehman Brothers Holdings Inc. filed for *Chapter 11 bankruptcy protection*.

Analysing the regime-1 probabilities in figure 2, we find that all our two-regime Markov-switching models generate several pronounced regime switches. The majority of these regime switches appear to occur at the same time, irrespective of the chosen Markov-switching specification. The most clear-cut example is the switch in November 2008, possibly indicating a structural change in German excess returns since the financial crisis.

It should be noted that in general, the timeline of the regime-1 probabilities estimated via the *ex-ante* and the smoothed probabilities of our Markov-switching GARCH models need not necessarily coincide with the low- and high-volatility periods depicted in the neighbouring panels. One reason might be that each Markov-switching GARCH specification allows for both a switching mean and a switching volatility equation, so that a regime-switch may be induced solely by a switch in the mean equation. A second explanation is that each regime-specific variance equation is capable of capturing certain qualitative volatility features (e.g. specific volatility asymmetries) which do not directly affect the volatility level, but which may nevertheless induce a structural switch from one regime to the other.

Finally, we reconsider the parameter estimates of the Free–Free model in table 4. Within this model class, we can test for pairwise equality of the corresponding regime-specific volatility parameters (i.e. $\mu_1 = \mu_2, \nu_1 = \nu_2, \omega_1 = \omega_2, \alpha_1 = \alpha_2, \beta_1 = \beta_2, b_1 = b_2, c_1 = c_2$). Apart from the parameter pairs (ν_1, ν_2) and (b_1, b_2), all other corresponding volatility parameters appear to be considerably different from each other across both regimes, thus indicating substantial structural differences between both volatility equations in the German stock index DAX.

3.3. *Out-of-sample volatility-forecasting performance*

So far, we have investigated the in-sample properties of our generalized Markov-switching GARCH approach. In this section, we adopt an approach outlined in Klaassen (2002) to assess the out-of-sample volatility-forecasting performance of the framework. For clarity of exposition, we start by analysing one-period-ahead variance forecasts and then discuss an extension to cover multiple-period-ahead volatility forecasts.

In what follows, we denote the one-period-ahead variance of the return, conditional on information available at time t by $\mathrm{Var}_t(r_{t,1})$. Using the recursive structure of our entire Markov-switching GARCH framework as presented in section 2, we obtain the expression

$$\begin{aligned}\mathrm{Var}_t(r_{t,1}) &= h_t \\ &= p_{1,t}\left(h_{1,t} + \left[\lambda_1 + \gamma_1\sqrt{h_{1,t}}\right]^2\right) \\ &\quad + (1-p_{1,t})\left(h_{2,t} + \left[\lambda_2 + \gamma_2\sqrt{h_{2,t}}\right]^2\right) \\ &\quad - \left[p_{1,t}\left(\lambda_1 + \gamma_1\sqrt{h_{1,t}}\right) + (1-p_{1,t})\left(\lambda_2 + \gamma_2\sqrt{h_{2,t}}\right)\right]^2. \quad (15)\end{aligned}$$

Based on this theoretical formula, we finally obtain our one-period-ahead variance forecasts $\widehat{\mathrm{Var}}_t(r_{t,1})$ by replacing all theoretical parameters inherent in the right of equation (15) by their respective ML estimates.

In order to implement our out-of-sample forecasting procedure, we use the parameter estimates from table 4, which were obtained from the entire sampling period (3 January 2000–31 December 2009). We generate our one-period-ahead volatility forecasts $\widehat{\mathrm{Var}}_t(r_{t,1})$ with additional data covering the daily excess returns between 4 January 2010 and 30 December 2010 (255 observations). Owing to the computational burden, we use a so-called fixed sampling scheme, that is, we do not update our estimates, but keep our parameters estimates fixed over the entire forecasting period.

To evaluate the quality of our one-period-ahead volatility forecasts, we have to compare them with the 'observed one-period-ahead volatility' which we denote by $\mathrm{ObsVar}_{t,1}$. In line with Andersen and Bollerslev (1998), we use a measure of observed volatility obtained from intra-daily data. Specifically, we define observed volatility as the sum of 15-min squared returns over each day.†

†The intra-daily data were provided by the *Karlsruher Kapitalmarktdatenbank (KKMDB)*. See also Marcucci (2005), who uses this approach in the context of volatility forecasting with Markov-switching GARCH models.

Based on the 'observed one-period-ahead volatility' $\text{ObsVar}_{t,1}$ and the one-period-ahead forecasts $\widehat{\text{Var}}_t(r_{t,1})$, we ultimately assess the forecasting quality of our Markov-switching GARCH specifications via the standard forecast efficiency regression

$$\text{ObsVar}_{t,1} = \psi_0 + \psi_1 \cdot \widehat{\text{Var}}_t(r_{t,1}) + \eta_{t+1}. \tag{16}$$

Table 6 shows the estimation results of the efficiency regression (16) for all 18 two-regime and single-regime specifications presented in table 3. Standard errors are corrected for autocorrelation and heteroscedasticity, according to the procedure proposed by Newey and West (1987). Moreover, using a correction method suggested by West and McCracken (1998), we also account for a potential impact on the standard errors that might arise from the fact that the parameters used to compute the forecasts are not known, but estimated from the data. In order to compare the volatility forecasts of our two-regime and single-regime GARCH specifications, table 6 also contains the R^2-type measure proposed in Klaassen (2002). This measure equals the standard R^2 of the forecast efficiency regression (16) computed under the restriction $\psi_0 = 0$ and $\psi_1 = 1$:

$$\tilde{R}^2 = 1 - \frac{\text{Var}\left[\text{ObsVar}_{t,1} - \widehat{\text{Var}}(r_{t,1})\right]}{\text{Var}\left[\text{ObsVar}_{t,1}\right]}. \tag{17}$$

Considering the two-regime Markov-switching GARCH models with the GARCH-in-Mean term in group (1), we find that for the one-period forecasting horizon, the $\tilde{R}^2$ statistics range between 0.0678 for the standard Markov-switching

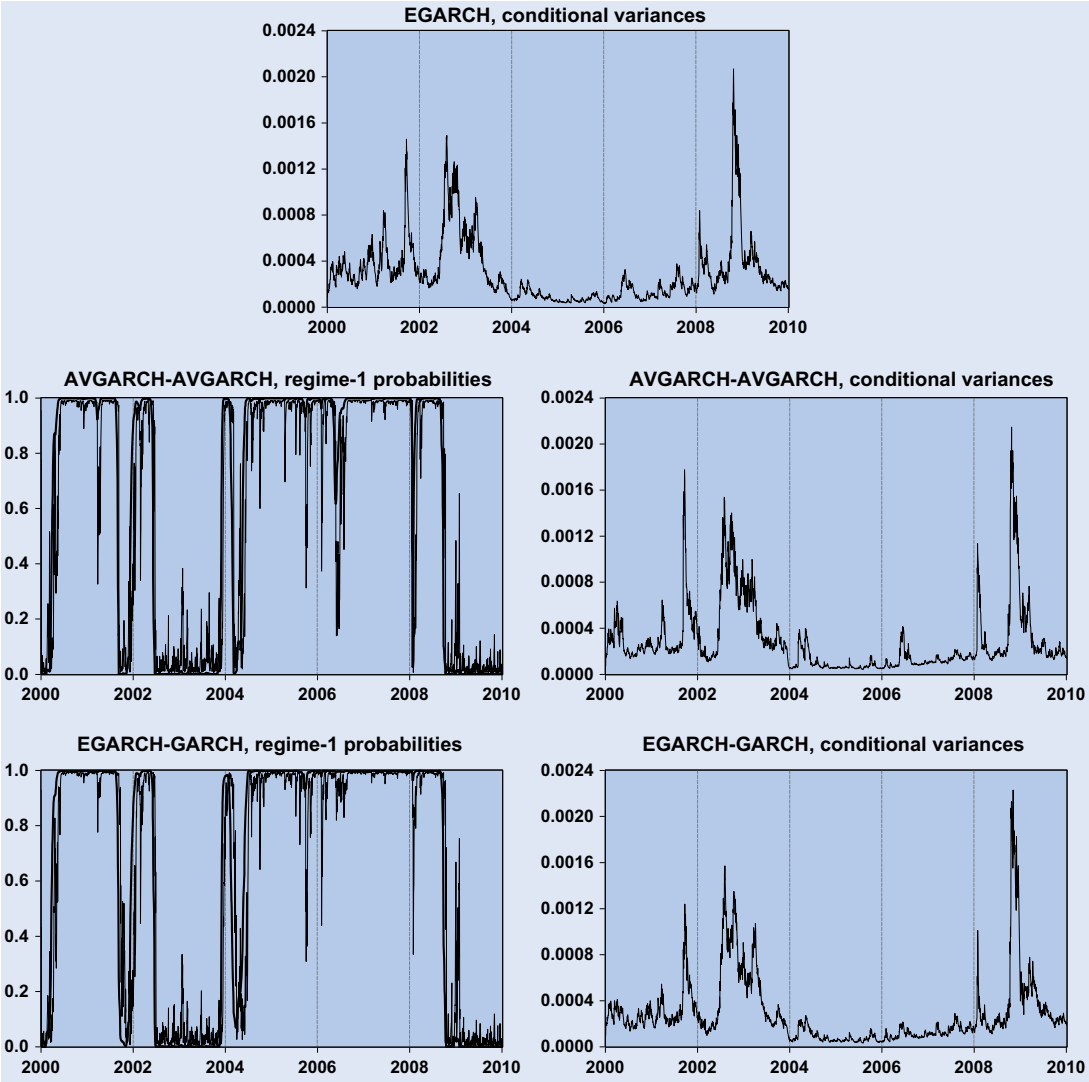


Figure 2. The respective panels display the ex-ante (thin lines) and the smoothed regime-1 probabilities (bold lines), as well as the conditional variances of the daily DAX excess returns as estimated via the single-regime EGARCH model (conditional variances only) and the two-regime Markov-switching AVGARCH–AVGARCH, EGARCH–GARCH, EGARCH–EGARCH and Free–Free specifications, respectively. All models display a quasi-uniform volatility pattern (represented by the conditional variances) during the sampling period. The German stock market started the decade with a period of relatively high volatility with a pronounced peak around 11th September 2001. After a short phase of normalization, high stock market volatility occurred between mid-2002 and mid-2003 reflecting the German bear market. After a phase of comparably low and moderate volatility between 2004 and 2008, the German stock market began to respond to the subprime crisis at the beginning of the year 2008 with a volatility peak around the Lehman Brothers bankruptcy on 15 September 2008.

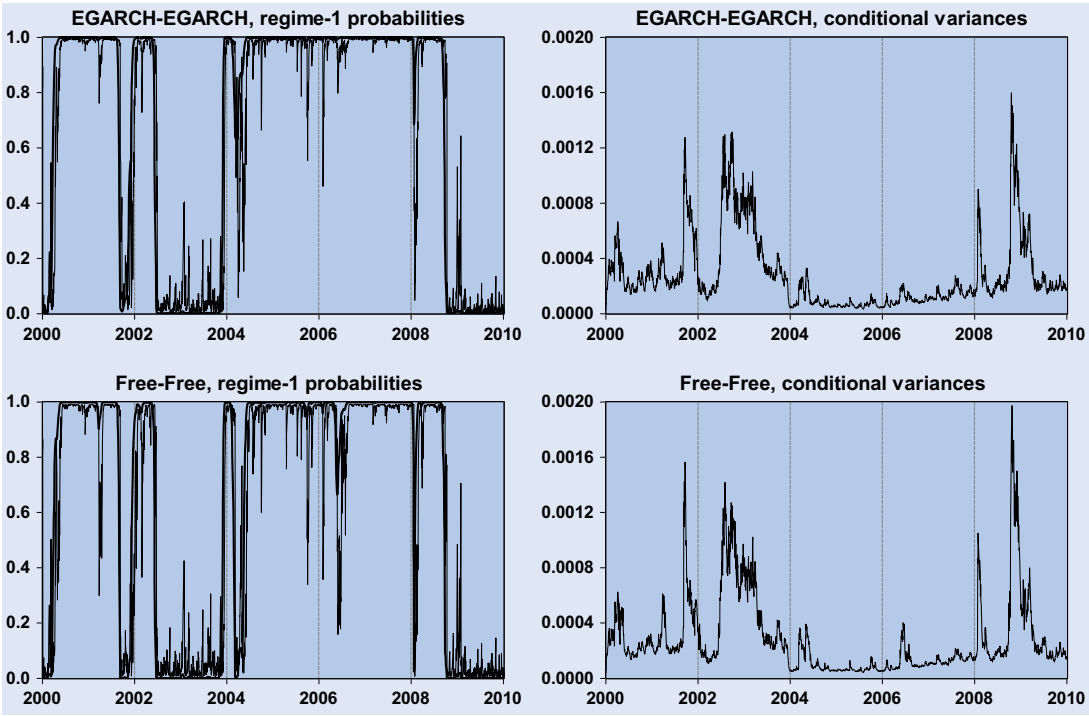


Figure 2. (Continued).

GARCH model and 0.3563 for our Free–Free specification. Overall, for the one-period forecast horizon, our analysis exhibits major relative outperformance of the Free–Free model over the EGARCH–GARCH and the EGARCH–EGARCH models, while the relative forecasting improvement over the AVGARCH–AVGARCH specification appears to be small.

Group (2) of table 6 contains the one-period horizon results for the two-regime Markov-switching GARCH models without the GARCH-in-Mean component. Again, the Free–Free specification relatively outperforms the alternative specifications with moderate improvements over the AVGARCH–AVGARCH and the EGARCH–EGARCH models. However, comparing the $\tilde{R}^2$ statistics of the specifications from group (2) with their counterpart specifications from group (1), we conclude that the inclusion of the GARCH-in-Mean term appears to improve out-of-sample forecasting performance. Groups (3) and (4) of table 6 contain the single-regime specifications with and without GARCH-in-Mean terms, respectively. Again, the inclusion of the GARCH-in-Mean component improves the one-period-ahead forecasting performance.

Finally, we show how to recursively use our framework to obtain volatility forecasts for multiple periods ahead. In contrast to Klaassen (2002), analytic expressions for multi-horizon volatility forecasts are difficult to obtain in our generalized framework. While Klaassen's recursive volatility-forecasting procedure makes extensive use of the fact that in his framework the one-period changes are free of serial correlation, our generalized Markov-switching GARCH set-up explicitly allows for such serial correlation. In view of this complication, we conduct a simulation-based recursive volatility-forecasting procedure for the specifically chosen forecast horizon of 10 periods (days) ahead.

Following the recursive volatility-forecasting procedure described in Klaassen (2002, p. 369), we also aim at using the conditional expected value of h_t from equation (15) in each recursion step. However, owing to non-linearities of h_t stemming from equations (6) to (8), this conditional expectation cannot be computed analytically. Therefore, we use a simulated expected value obtained from 10 000 future returns sampled according to equation (2) in each recursion step.

The right half of table 6 presents the estimation results of the efficiency regression (16), along with the $\tilde{R}^2$ statistics of the 18 two-regime and single-regime GARCH specifications for the 10-period horizon. Again, the Free–Free GARCH-in-Mean specification in group (1) slightly outperforms the AVGARCH–AVGARCH model and yields more pronounced forecasting improvements than all other two-regime GARCH-in-Mean specifications. Overall, the qualitative nature of the forecasting results for the 10-period horizon across and within the four specification groups appears to be quite similar to that obtained for the one-period horizon given in the left half of table 6.

4. Summary and conclusion

In this paper, we establish a two-regime Markov-switching GARCH model which enables us to estimate complex functional GARCH specifications within each regime. Combining Gray's (1996) and Klaassen's (2002) Markov-switching framework with Hentschel's (1995) approach of nesting alternative single-regime GARCH models, our framework unifies many commonly used Markov-switching GARCH models. Our theoretical set-up also features two further components discussed in the literature, namely (a) a GARCH-in-Mean term in the mean equation and (b) time-varying transition probabilities governing the dynamics of the latent regime indicator.

Table 6. Estimates and statistics of the standard forecast efficiency regression (two-regime and single-regime GARCH specifications).

	One-period horizon			Ten-period horizon		
Model	ψ_0	ψ_1	$\tilde{R}^2$	ψ_0	ψ_1	$\tilde{R}^2$
(1) GARCH-in-Mean:						
GARCH-GARCH	5.64×10^{-5}	1.2374	0.0678	−0.0002	1.8347	0.0094
	(1.47×10^{-5})	(0.4558)		(0.0002)	(1.0982)	
AVGARCH-AVGARCH	-9.86×10^{-5}	0.9402	0.3513	6.32×10^{-6}	1.2524	0.1832
	(2.67×10^{-5})	(0.1441)		(1.61×10^{-5})	(0.2091)	
EGARCH-GARCH	-3.48×10^{-5}	0.5595	0.0948	-3.77×10^{-5}	0.7408	0.0459
	(2.78×10^{-5})	(0.1329)		(4.93×10^{-5})	(0.2912)	
EGARCH-EGARCH	−0.0001	1.1691	0.2378	3.42×10^{-5}	0.8099	0.1276
	(2.68×10^{-5})	(0.1449)		(1.33×10^{-5})	(0.1735)	
Free-Free	-9.63×10^{-5}	1.0197	0.3563	4.49×10^{-6}	1.3408	0.1852
	(2.88×10^{-5})	(0.1683)		(1.62×10^{-5})	(0.2153)	
(2) No GARCH-in-Mean:						
GARCH-GARCH	1.84×10^{-5}	0.6201	0.0539	0.0001	0.5303	0.0045
	(2.15×10^{-5})	(0.1722)		(1.71×10^{-5})	(0.0442)	
AVGARCH-AVGARCH	1.96×10^{-5}	1.1311	0.2319	-2.13×10^{-5}	0.5002	0.1331
	(1.14×10^{-5})	(0.1244)		(2.16×10^{-5})	(0.1052)	
EGARCH-GARCH	1.59×10^{-5}	0.6569	0.0653	0.0002	0.6605	0.0022
	(2.13×10^{-5})	(0.1766)		(0.0002)	(0.9090)	
EGARCH-EGARCH	2.09×10^{-5}	1.2350	0.2264	1.09×10^{-5}	1.1102	0.1225
	(1.34×10^{-5})	(0.1925)		(1.84×10^{-5})	(0.2528)	
Free-Free	3.87×10^{-5}	0.9597	0.2560	-2.96×10^{-5}	0.8895	0.1628
	(9.49×10^{-6})	(0.0960)		(3.14×10^{-5})	(0.1229)	
(3) GARCH-in-Mean:						
GARCH	0.0002	0.2576	0.0315	0.0004	1.1457	0.0044
	(2.22×10^{-5})	(0.0791)		(0.0003)	(1.0306)	
AVGARCH	1.30×10^{-5}	1.1156	0.1307	-5.31×10^{-5}	1.4328	0.0848
	(1.68×10^{-5})	(0.2388)		(3.84×10^{-5})	(0.4021)	
EGARCH	2.34×10^{-5}	0.8333	0.1861	−0.0001	0.8513	0.1049
	(1.41×10^{-5})	(0.1323)		(6.34×10^{-5})	(1.1134)	
Free	2.28×10^{-5}	1.2413	0.2303	1.79×10^{-5}	0.8542	0.1133
	(1.29×10^{-5})	(0.1902)		(1.72×10^{-5})	(0.1914)	
(4) No GARCH-in-Mean:						
GARCH	−0.0002	1.7319	0.0070	9.75×10^{-5}	0.6161	0.0016
	(0.0002)	(1.1119)		(1.52×10^{-5})	(0.0282)	
AVGARCH	0.0005	1.3492	0.1171	-2.78×10^{-5}	1.0546	0.0805
	(9.66×10^{-5})	(0.4125)		(3.19×10^{-5})	(0.2997)	
EGARCH	−0.0003	1.5605	0.1520	-4.29×10^{-7}	1.0119	0.1031
	(7.79×10^{-5})	(0.5290)		(2.30×10^{-5})	(0.2515)	
Free	2.34×10^{-5}	0.8565	0.1795	3.84×10^{-6}	1.0811	0.1083
	(1.41×10^{-5})	(0.1374)		(2.32×10^{-5})	(0.2644)	

Notes: Estimates of the parameters from equation (16). Standard errors are in parentheses. $\tilde{R}^2$ denotes the forecasting statistic from equation (17).

In the technical appendix to the paper, we develop a reliable maximum-likelihood estimation algorithm. In the empirical part, we apply the estimation procedure to daily excess returns of the German stock index DAX from the preceding decade. We evaluate the outperformance of our most general Markov-switching Free–Free model over a set of nested specifications in terms of statistical fit and out-of-sample volatility forecasting performance. As a major empirical result, we find significant Markov-switching structures in German stock-market data, with substantially differing volatility equations across both Markov-regimes.

Finally, we note that our Markov-switching GARCH framework assumes Gaussian innovations in equations (1) and (2). However, several authors successfully apply alternative distributions in Markov-switching GARCH models. Klaassen (2002) and Ardia (2009), for example, argue that Student-t innovations within the regimes are capable of enhancing the stability of the states (see also Bulla 2011). In view of this, a useful line of future research could entail the extension of our framework to more general innovation distributions.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendix 1. Maximum likelihood estimation

In this appendix, we construct the log-likelihood function of our Markov-switching GARCH model established in section 2. We only consider the two-regime case, although a theoretical extension of the entire framework to more Markov regimes is straightforward.

The conditional probability distribution of r_{t+1} is shown in equation (2). The corresponding probability density function has the form

$$\begin{aligned} f(r_{t+1}|\phi_t) &= \sum_{i=1}^{2} f(r_{t+1}, S_t = i|\phi_t) \\ &= \sum_{i=1}^{2} \Pr(S_t = i|\phi_t) \cdot f(r_{t+1}|S_t = i, \phi_t) \\ &= \sum_{i=1}^{2} p_{i,t} \cdot f(r_{t+1}|S_t = i, \phi_t), \end{aligned} \tag{A1}$$

where, as in the main text, $p_{i,t} \equiv \Pr(S_t = i|\phi_t)$ denotes the *ex-ante* regime-i probability. The information set ϕ_t consists of the entire history of $\tilde{r}_t = \{r_t, r_{t-1}, \ldots\}$.

Since the regime indicator S_t follows a first-order Markov process, the *ex-ante* probability $p_{i,t}$ depends only on S_{t-1} and r_t. Using the *Theorem of Total Probabilities*, we obtain

$$p_{i,t} = \sum_{j=1}^{2} \Pr(S_t = i|S_{t-1} = j, \tilde{r}_t) \Pr(S_{t-1} = j|\tilde{r}_t). \tag{A2}$$

The first probability $\Pr(S_t = i|S_{t-1} = j, \tilde{r}_t)$ on the right-hand side of (A2) does not depend on the entire history of $\tilde{r}_t$, so that we replace $\tilde{r}_t$ by r_t in this latter probability. Thus, we can insert the probabilities specified in equation (9) in equation (A2) and obtain

$$\begin{aligned} p_{1,t} &= P_t \cdot \Pr(S_{t-1} = 1|\tilde{r}_t) + (1 - Q_t) \cdot \Pr(S_{t-1} = 2|\tilde{r}_t) \\ &= P_t \cdot \Pr(S_{t-1} = 1|\tilde{r}_t) + (1 - Q_t) \cdot (1 - \Pr(S_{t-1} = 1|\tilde{r}_t)), \end{aligned} \tag{A3}$$

and analogously

$$p_{2,t} = Q_t \cdot (1 - \Pr(S_{t-1} = 1|\tilde{r}_t)) + (1 - P_t) \cdot \Pr(S_{t-1} = 1|\tilde{r}_t). \tag{A4}$$

The remaining probability $\Pr(S_{t-1} = 1|\tilde{r}_t)$ in equations (A3) and (A4) can be written as a function of $p_{1,t-1} = \Pr(S_{t-1} = 1|\tilde{r}_{t-1})$. To this end, we apply *Bayes' Formula*, yielding

$$\begin{aligned} \Pr(S_{t-1} = 1|\tilde{r}_t) &= \Pr(S_{t-1} = 1|r_t, \tilde{r}_{t-1}) \\ &= \frac{f(r_t|S_{t-1} = 1, \tilde{r}_{t-1}) \Pr(S_{t-1} = 1, \tilde{r}_{t-1})}{\sum_{i=1}^{2} f(r_t|S_{t-1} = i, \tilde{r}_{t-1}) \Pr(S_{t-1} = i, \tilde{r}_{t-1})}. \end{aligned} \tag{A5}$$

Expanding the ratio on the right-hand side of equation (A5), we obtain

$$\begin{aligned} \Pr(S_{t-1} = 1|\tilde{r}_t) &= \frac{f(r_t|S_{t-1} = 1, \tilde{r}_{t-1}) \Pr(S_{t-1} = 1|\tilde{r}_{t-1})}{\sum_{i=1}^{2} f(r_t|S_{t-1} = i, \tilde{r}_{t-1}) \Pr(S_{t-1} = i|\tilde{r}_{t-1})} \\ &= \frac{f(r_t|S_{t-1} = 1, \tilde{r}_{t-1}) p_{1,t-1}}{\sum_{i=1}^{2} f(r_t|S_{t-1} = i, \tilde{r}_{t-1}) p_{i,t-1}} \\ &= \frac{g_{1,t-1} \cdot p_{1,t-1}}{\sum_{i=1}^{2} g_{i,t-1} \cdot p_{i,t-1}}, \end{aligned} \tag{A6}$$

where, for ease of notation, we have defined $g_{i,t-1} \equiv f(r_t|S_{t-1} = i, \tilde{r}_{t-1}) = f(r_t|S_{t-1} = i, \phi_{t-1})$. Using equation (A6), we are now able to calculate the *ex-ante* probability $p_{1,t}$ by inserting equation (A6) in equation (A3):

$$\begin{aligned} p_{1,t} &= P_t \cdot \frac{g_{1,t-1} p_{1,t-1}}{g_{1,t-1} p_{1,t-1} + g_{2,t-1}(1 - p_{1,t-1})} \\ &\quad + (1 - Q_t) \cdot \left[1 - \frac{g_{1,t-1} p_{1,t-1}}{g_{1,t-1} p_{1,t-1} + g_{2,t-1}(1 - p_{1,t-1})} \right] \\ &= P_t \cdot \frac{g_{1,t-1} p_{1,t-1}}{g_{1,t-1} p_{1,t-1} + g_{2,t-1}(1 - p_{1,t-1})} \\ &\quad + (1 - Q_t) \cdot \frac{g_{2,t-1}(1 - p_{1,t-1})}{g_{1,t-1} p_{1,t-1} + g_{2,t-1}(1 - p_{1,t-1})}. \end{aligned} \tag{A7}$$

Next, we address the exact form of the conditional density f appearing in equations (A1) and (A7). As we are assuming conditional normality, f is given as follows:

$$\begin{aligned} &f(r_{t+1}|S_t = i, \phi_t) \\ &= \frac{1}{\sqrt{2\pi h_{i,t}}} \exp\left\{ -\frac{\left[r_{t+1} - (\lambda_i + \gamma_i \sqrt{h_{i,t}}) \right]^2}{2 h_{i,t}} \right\}. \end{aligned} \tag{A8}$$

The variance $h_{i,t}$ depends on the explicit functional form of the GARCH equation. It is easy to verify from equation (8) that it can be written as

$$h_{i,t} = \begin{cases} \left[\omega_i + \alpha_i \sqrt{h_{t-1}^{(i)}}^{\mu_i} f_i^{\nu_i}(\delta_t^{(i)}) + \beta_i \sqrt{h_{t-1}^{(i)}}^{\mu_i} \right]^{2/\mu_i} & \text{for } \mu_i > 0 \\ \left[\exp\left\{ \omega_i + \alpha_i f_i^{\nu_i}(\delta_t^{(i)}) + \beta_i \ln\left(\sqrt{h_{t-1}^{(i)}} \right) \right\} \right]^2 & \text{for } \mu_i = 0 \end{cases}, \tag{A9}$$

with appropriately defined parameters $\omega_i, \alpha_i, \beta_i$.

It is obvious from equation (A9) that for the calculation of regime-specific variances $h_{i,t}$, we need the aggregated variances and shock terms $h_{t-1}^{(i)}$ and $\delta_t^{(i)}$ the calculation of which we base on the Klaassen (2002) probabilities $p_{1,t-1}^{(i)}$ as described in the main text. Using *Bayes' Formula* again, we obtain the Klaassen probabilities as

$$\begin{aligned} p_{1,t-1}^{(i)} &= \Pr(S_{t-1} = 1|\tilde{r}_t, S_t = i) \\ &= \frac{\Pr(S_t = i|\tilde{r}_t, S_{t-1} = 1) \Pr(S_{t-1} = 1|\tilde{r}_t)}{\Pr(S_t = i|\tilde{r}_t)} \\ &= \frac{\Pr(S_t = i|\tilde{r}_t, S_{t-1} = 1) \cdot p_{1,t-1}}{\Pr(S_t = i|\tilde{r}_t)}, \end{aligned} \tag{A10}$$

with $p_{1,t-1}$ as given in equation (A7). Applying the *Theorem of Total Probabilities* once more, we write the denominator in equation (A10) as

$$\begin{aligned} \Pr(S_t = i|\tilde{r}_t) &= \Pr(S_t = i|\tilde{r}_t, S_{t-1} = 1) \cdot p_{1,t-1} \\ &\quad + \Pr(S_t = i|\tilde{r}_t, S_{t-1} = 2) \cdot (1 - p_{1,t-1}). \end{aligned} \tag{A11}$$

To calculate the probability on the left of equation (A11), we need the two probabilities $\Pr(S_t = i|\tilde{r}_t, S_{t-1} = 1)$ and $\Pr(S_t = i|\tilde{r}_t, S_{t-1} = 2)$, which coincide with the probabilities given in the equations (9) and (10) in the main text. Now, inserting these probabilities and (A11) into equation (A10), we obtain

$$\begin{aligned} p_{1,t-1}^{(1)} &= \frac{\Phi(d_1 + e_1 \cdot r_t) p_{1,t-1}}{\Phi(d_1 + e_1 \cdot r_t) p_{1,t-1} + \{1 - \Phi(d_2 + e_2 \cdot r_t)\} p_{2,t-1}}, \\ p_{2,t-1}^{(1)} &= 1 - p_{1,t-1}^{(1)}, \\ p_{1,t-1}^{(2)} &= \frac{\{1 - \Phi(d_1 + e_1 \cdot r_t)\} p_{1,t-1}}{\{1 - \Phi(d_1 + e_1 \cdot r_t)\} p_{1,t-1} + \Phi(d_2 + e_2 \cdot r_t) p_{2,t-1}}, \\ p_{2,t-1}^{(2)} &= 1 - p_{1,t-1}^{(2)}. \end{aligned} \tag{A12}$$

Finally, we use the recursive structures developed so far to construct the log-likelihood function of our flexible Markov-switching model defined in the equations (1)–(10). The general form of the likelihood function is

$$L(\boldsymbol{\Theta}) = f(r_t, \ldots, r_1; \boldsymbol{\Theta}),$$

with the vector $\boldsymbol{\Theta}$ containing all model parameters. Writing this joint distribution of the returns as a product of conditional densities, we obtain

$$L(\boldsymbol{\Theta}) = \prod_{t=1}^{T} f(r_t|\tilde{r}_{t-1}; \boldsymbol{\Theta}),$$

for which we define the starting term as $f(r_1|\tilde{r}_0; \boldsymbol{\Theta}) \equiv f(r_1; \boldsymbol{\Theta})$. Taking the logarithm of $L(\boldsymbol{\Theta})$ and inserting (the lagged form of) equation (A1), we obtain the log-likelihood function as

$$\begin{aligned}\ell(\boldsymbol{\Theta}) &\equiv \log[L(\boldsymbol{\Theta})] \\ &= \sum_{t=1}^{T} \log\left[f(r_t|\tilde{r}_{t-1}; \boldsymbol{\Theta})\right] \\ &= \sum_{t=1}^{T} \log\left[\sum_{j=1}^{2} f(r_t|S_{t-1} = j, \tilde{r}_{t-1}; \boldsymbol{\Theta}) \cdot p_{j,t-1}\right]. \quad \text{(A13)}\end{aligned}$$